

Three Ways to Miss a Grasp: A Same-Metric Comparison of Rule-Based, Learned, and Zero-Shot Vision-Language Grasp Prediction, and a Test of Why the Language Model Fails

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<https://github.com/anishtalla27/Vision-Based-Grasping>

Abstract

A robot cannot grasp an object until something decides where to place the fingers and at what angle. That decision can come from hand-written rules, from a network trained on labeled grasps, or from a general-purpose vision-language model never trained on grasping, and the three are rarely compared under one protocol. This paper scores one system of each kind with a single implementation of the Cornell rectangle metric on the same held-out split of 123 images of 35 unseen objects, with object-clustered statistics and a shared failure taxonomy. A fine-tuned ResNet34 passed on 84.0% of test images (three-seed mean), a detector-plus-lookup-table heuristic on a post-hoc-corrected 57.7%, and zero-shot GPT-4o on 12.4%. The systems also failed in different ways, and GPT-4o’s errors suggested that it could identify a grasp but not express it as pixel coordinates. A follow-up experiment tested that account by asking the same model to choose among label-free candidate grasps drawn on the image. Its accuracy stayed at the random floor on three candidate menus, and it consistently preferred a pinch along the object’s long axis. Removing the coordinate output does not recover performance, which points to the grasp choice itself, although difficulty interpreting the drawn marks cannot yet be excluded.

Keywords

robotic grasping; Cornell Grasping Dataset; convolutional neural networks; vision-language models; GPT-4o; Set-of-Mark prompting; zero-shot evaluation; object-clustered bootstrap; failure analysis; spatial grounding

1 Introduction

Before a gripper closes on anything, something has to choose a contact point, an approach angle, and how wide to open the fingers. Every subsequent motion depends on that choice being approximately

correct. A mechanically capable arm will still fail often if the component that selects the grasp is wrong, so grasp prediction is a stage at which a small difference in accuracy produces a large difference in behavior. The decision must also be made from limited information. In most practical settings the robot has a camera image but no model of the object and no label identifying it.

Three broadly different approaches exist, and they are not variations on a single method. A programmer can encode rules directly, for example that a mug is grasped by its handle and a bottle around its neck. A model can be trained on thousands of hand-labeled examples until it learns to map pixels to grasp rectangles. Alternatively, a general-purpose vision-language model, trained on web text and images and never shown a grasping dataset, can be asked in natural language where to grip. Each approach places the relevant knowledge in a different part of the perception stack, whether explicit rules, weights fitted to task-specific data, or the general knowledge of a large model.

The third option is recent, and the field has not settled how to use it. Foundation models entered robotics quickly, and the literature has not yet established which parts of a manipulation pipeline they should handle. The question matters practically, because the three approaches demand very different resources. A lookup table needs only hand-specified rules, and a learned predictor needs a task-specific annotated dataset and the compute to train on it, whereas prompting a hosted vision-language model requires only an API key and minimal setup. Because that convenience makes the approach attractive, its accuracy should be measured before it is relied on.

Comparisons across those three families are rare. Most published grasp-detection work benchmarks within a family, comparing one trained architecture against another on the same dataset. Such benchmarks are useful, but they do not show what is gained or lost by changing approach entirely. Rule-based baselines, when they appear, are often described but not scored. Vision-language models usually appear embedded inside a larger system, where their individual contribution cannot be isolated. This study evaluates all three on the same test images, with a common output format and a single scoring implementation.

This paper asks which approach works and where each one breaks down. The second question is treated as equal to the first, because an accuracy figure shows only that a system was wrong, while the pattern of its errors points to a cause that a designer can act on.

The evaluation protocol is adopted from prior work. A grasp here is a rectangle with a center point, an orientation, and a width matched to how far the gripper opens, the representation Jiang et al. [1] introduced with the Cornell Grasping Dataset. Each image has several labeled grasps, since most objects can be picked up in more than one valid way, and a prediction counts as correct if it matches any of them. The matching rule is that the predicted angle falls within 30 degrees of a labeled grasp and the intersection over union between the two rectangles exceeds 25%, the convention established by Lenz et al. [2]. An external ground truth was used because an earlier design, in which the author would have written the ground truth and then graded a rule-based system against it, was close to circular. The dataset’s existing annotations avoid that problem and make the results comparable to published ones.

Four systems are evaluated. System A is the rule-based baseline, in which a COCO-pretrained detector identifies the object and a small fixed table maps the detected category to a

grasp region, with a geometric fallback covering the images the detector cannot name. System B is the learned predictor, a small convolutional network built from scratch alongside fine-tuned ResNet18 and ResNet34 models, trained on the Cornell training split. System C is GPT-4o, prompted directly with the raw image and given no task-specific training. System D is the same model shown a menu of candidate grasps drawn on the image and asked only to choose one, a follow-up run added after the first three systems were scored, to test the explanation their results suggested.

This paper does not attempt to improve on state-of-the-art grasp detection, and published pixel-wise systems reach higher accuracies than any system evaluated here. Its aim is to measure, under conditions held as close to identical as the design allows, how three families of technique fail, and to show that the pattern of each failure contains information that the accuracy figure does not.

This paper makes the following contributions.

1. A same-metric comparison of a rule-based, a learned, and a zero-shot vision-language grasp predictor, run on the same 123 held-out images of unseen objects through one shared scoring path.
2. Uncertainty reported at the level of the object, the unit that matches the stated population of unseen objects. The ordering of the three systems is preserved under an object-clustered bootstrap and permutation test, and the learned system is reported over three seeds under selection rules fixed before the test split was reopened.
3. A failure taxonomy and a center-hull diagnostic that separate the three families along lines a single accuracy number does not show.
4. A marked-candidate experiment (System D) with a floor and a ceiling computed on its own menu. It tests the most natural explanation of the vision-language model’s failure, that the model cannot express a chosen grasp as coordinates, and shows that removing the coordinate output is not sufficient to recover performance.
5. A public release of the code, the object-wise split, and every raw model reply.

1.1 Relation to prior work

This study builds on existing components: the Cornell Grasping Dataset and its rectangle representation [1], the 30-degree and 25% IoU pass rule [2], the ResNet architectures with ImageNet weights [6], Faster R-CNN with COCO weights [7], the sine and cosine encoding of twice the grasp angle [3, 4], GPT-4o [8], and marked-candidate prompting [9]. The work specific to this study is the shared evaluation pipeline, the reconstruction of object identities and the object-wise split, the detector-plus-table baseline, the training recipe and pre-registered second round of the learned system, the five-repeat protocol for the zero-shot model, the failure diagnostics, the object-clustered statistical analysis, and the candidate generator and menus that make up System D.

2 Methods

Fig. 1 summarizes the design. Every system receives the same RGB test image and must return the same kind of output, a grasp rectangle, which is scored by one function against the dataset’s labeled grasps. This section describes the design only. All numbers are reported in Section 3 and interpreted in Section 4.

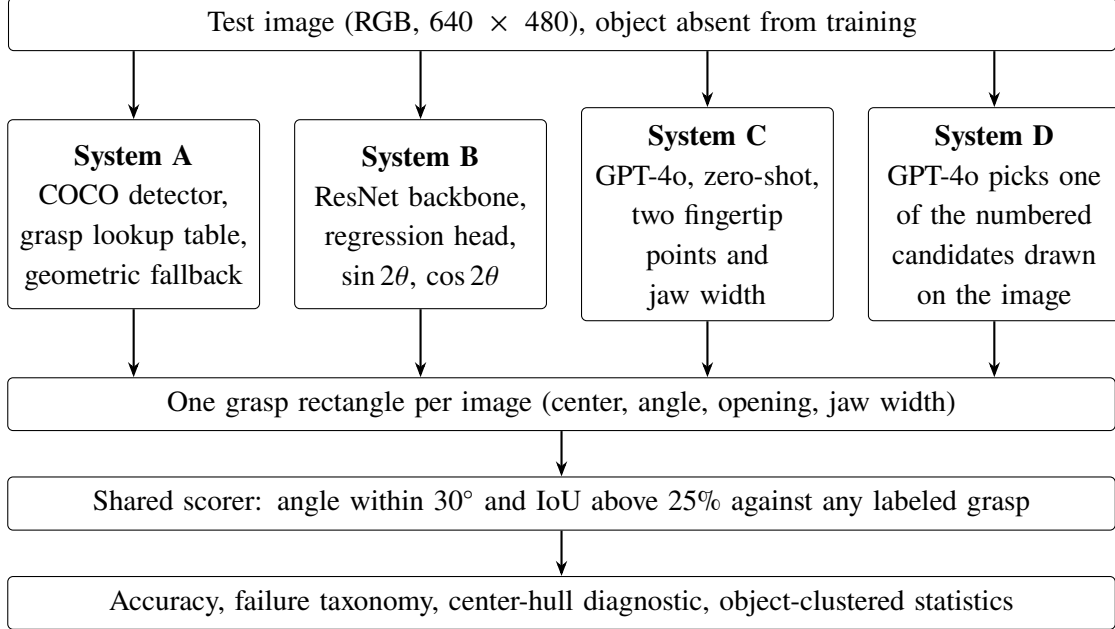


Fig. 1. Overview of the comparison. The four systems differ only in how the grasp rectangle is produced. Everything below the system row is shared.

2.1 Dataset and evaluation

The dataset used for evaluation is the Cornell Grasping Dataset, roughly a thousand images of household objects, each with hand-labeled grasp rectangles [1]. A prediction counts as correct when its angle falls within 30 degrees of some labeled grasp and its rectangle intersection over union with that grasp exceeds 25%, the convention established by Lenz et al. [2]. Cornell’s annotations are not exhaustive, so passing this metric shows agreement with a labeled grasp and does not establish that the grasp would physically succeed. The Jacquard dataset was considered as an alternative, but its full distribution requires a signed agreement and an approved non-free-email request. Reconstructing object identity was the most difficult step in building the split. No object may appear in both training and test, or a model can memorize its grasp rectangles from one split and be credited for recognizing the same object in the other. Cornell provides no object-identity labels, so identity had to be reconstructed from the image sequence. Comparing raw pixel differences between consecutive frames failed, because the dataset deliberately rotates each object between shots and same-object and different-object scores overlapped. Instead, each frame was segmented against the photography platform, and the resulting object regions were compared to decide whether two

frames showed the same object or different ones.

Acceptance thresholds were set asymmetrically, since merging two different objects into one group is less harmful than splitting one real object into two. The “different-object” threshold was set below the lowest score seen for any confirmed same-object pair, while the “same-object” threshold was left permissive. A 30-boundary stratified sample was classified by both an AI assistant and a human, blind to each other’s calls. The AI’s classifications matched the human’s on 86.7% of its confident calls but only 40.0% of the calls it flagged as uncertain, so every uncertain call and every confident “different” call was reviewed by hand. Confident “same” calls were accepted unreviewed, since an error in that direction is harmless. The result is 234 objects across 883 images, with 620 for training, 140 for validation, and 123 for test, as shown in Table 1. Published descriptions of Cornell normally report around 240 objects, and the shortfall here is expected, since the thresholds were deliberately biased toward merging. Every merge that should have been a split lowers the object count while keeping the merged group on one side of the division, which is the direction of error this design deliberately accepts. Each object has at least one grasp rectangle, and most have several, since most objects can be picked up more than one way.

2.2 System A, rule-based baseline

System A is better described as a detector-assisted heuristic baseline than a purely rule-based one. It detects the object with a COCO-pretrained detector, which is itself a learned component, then looks up the detected category in a fixed table of grasps. No part of it is learned from grasp labels, which is the sense in which it is rule-based. The table was committed before any evaluation code existed, so no entry could be tuned toward a score. Since COCO’s categories miss more than half of what Cornell photographs, a geometric fallback that does not rely on object recognition covers the objects the detector misses, and that fallback handles the majority of test images.

System A was scored once before the detector was found to be drawing boxes around background clutter. The guard that corrected this, a requirement that the detected box contain the object’s centroid together with one constant recalibrated on training data, was added after those test predictions had been inspected. The decision to add it was therefore prompted by test behavior, and the corrected figure is a post-hoc result. This weakens the comparisons that rest on it, including the McNemar test against System B and the interval comparison against System C. Both the original and the corrected figure are reported wherever the result appears (Section 3.3).

COCO’s 80 categories do not cover most of the objects in Cornell, so many objects have no table entry. In addition, on either path System A’s rectangle orientation is limited to 0 or 90 degrees, the only two orientations the table and the box-based rule can express, so it cannot produce a grasp that requires an intermediate angle.

2.3 System B, learned grasp predictor

System B regresses a grasp rectangle directly from an image. A backbone network extracts features, pools them, and a regression head predicts the rectangle’s parameters. Orientation is encoded as

the sine and cosine of twice the angle rather than a raw angle between 0 and 180 degrees. This accounts for the fact that a grasp rectangle rotated 180 degrees describes the same physical grasp. A raw-angle representation would score that rotation as an entirely different grasp, while the sine and cosine encoding treats it as identical. Redmon and Angelova [3] and Morrison et al. [4] both used this same encoding.

System B trained three architectures on the same training split, a small network built from scratch, a fine-tuned ResNet18, and a fine-tuned ResNet34. The loss was computed against every labeled grasp on an image, with only the best match backpropagated. Averaging over all of an image’s grasps would place the target between, for example, a handle grasp and a rim grasp, at a point where neither is valid. Augmentation applied the same affine transform to the grasp rectangle’s corners as to the image’s pixels, instead of hand-writing a rule for how the angle behaves under a flip.

System B was trained twice. The first round trained each architecture once, at one seed, with a constant learning rate, and kept whichever of roughly a hundred epochs scored highest on validation. Validation accuracy on 140 images varies substantially between consecutive epochs (Section 3.4), so the maximum over epochs is a noisy selection criterion and a single run cannot separate one architecture from another. The second round kept the model, loss, data, metric, and split, and changed only the recipe: a cosine learning-rate decay over a fixed 100 epochs, an exponential moving average of the weights, three seeds per architecture, and no checkpoint selection at all, since the reported model is the final-epoch averaged weights. Test-time augmentation, eight dihedral views of each image with the prediction carried back through the same corner transform used in training and the medoid rectangle kept, was added and its use decided on validation. Four rules were fixed before the second round was evaluated on the test split: the architecture is the one with the higher mean final-weight validation accuracy across seeds, augmentation is used only if it raises that mean, the primary result is the mean over seeds of the final weights, and every configuration evaluated on the test split is reported. A 60-epoch budget was tried first and rejected on validation alone, since both architectures were still improving when the decay ended. Opening the test split a second time departs from the original single-opening plan.

2.4 System C, zero-shot vision-language model

System C prompts GPT-4o with each test image and asks where the two gripper fingertips should touch, with no training or fine-tuning on grasping. The prompt was developed on a batch of 30 training images and frozen before test was opened.

The model was called as `openai/gpt-4o` through the OpenRouter API, at a 400-token cap, with temperature left unset so the provider default applied and no seed pinned, since run-to-run variation was one of the quantities being measured. Images were sent at the dataset’s native 640 by 480 resolution with no resizing, and the prompt told the model that coordinates were in pixels of that image. Each repeat was an independent single-turn completion with no prior messages, so the five repeats of an image are independent draws. Transport failures were retried up to three times, but a reply that arrived and then failed to parse was never retried, since resampling malformed replies

would implicitly convert a single-shot baseline into a best-of-N one.

System C was not asked to output an angle. It returned two fingertip points and a jaw width as JSON, and those three values determine all four rectangle corners, which were then passed through the same frozen function that produced every ground-truth angle in the dataset. No rectangle parameter was supplied manually, and no angle convention had to be matched. Each image was prompted five times, and all five results are reported.

The design isolates one step that published systems usually avoid. In several grasping systems built on the same kind of model, the model’s output is a choice, a label, an order of operations, or an image, and the pose of the gripper comes from something else. FreeGrasp asks the model to decide which object to grasp and in what order, using marks placed on the image [11]. Lan-grasp asks the model which part to grasp, then hands the pose to a conventional planner [12]. ThinkGrasp uses the model to plan clutter removal [13]. Set-of-Mark asks the model to select one of several labeled regions [9]. VLAD-Grasp asks the model to draw the grasp using a virtual gripper [10]. In none of them does the model emit the pose coordinates itself. System C asks for exactly that, so that any gap can then be examined as a failure of coordinate emission or as a failure to perceive the object.

2.5 System D, marked-candidate selection

System D tests the coordinate-binding account directly. If GPT-4o can see where to grasp but cannot express it as pixel coordinates, then removing the coordinate output should recover its choice. The same model was therefore shown the same test images with twelve candidate grasps drawn on a zoomed crop of the object and asked to return the number of the one it would use. In this setting the model never emits a coordinate.

The candidates are generated without reading any grasp label. The platform segmentation already used for the object-wise split gives the object’s mask, and principal component analysis of that mask gives its centroid, long axis, and extent. Candidates are three positions along the long axis (the centroid and 35% of the extent to either side) crossed with four jaw-travel directions 45 degrees apart, starting from the short axis, so every possible grasp angle lies within 22.5 degrees of some candidate, inside the metric’s 30 degree tolerance. Opening and jaw width use System A’s train-calibrated constants. Each candidate is drawn as two thick fingertip pads joined by a thin line with inward arrowheads, in its own color, with a numbered badge. The numbering is re-shuffled on every call, so a preference for a particular number cannot be mistaken for a preference for a grasp, and self-agreement is measured on the physical candidate. Eleven test images have no segmentation mask and so no candidates. They count as misses for System D and for every reference row it is compared with, so the comparisons stay paired.

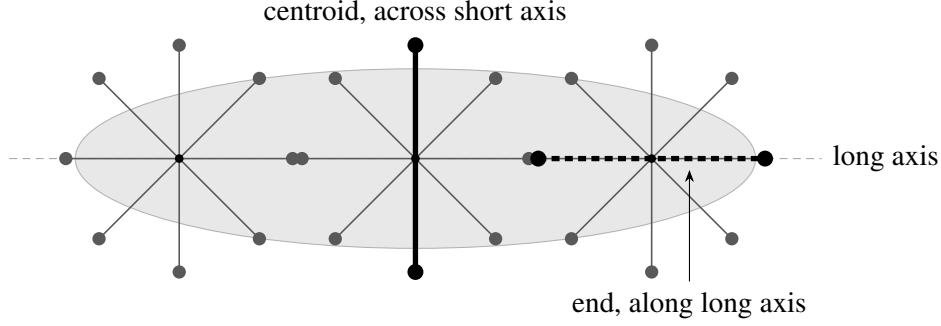


Fig. 2. Schematic of the twelve-candidate menu, not to scale. The object mask (gray) gives a centroid and a long axis. Candidates are three positions along that axis crossed with four jaw-travel directions, and each line joins the two fingertip pads of one candidate. The solid candidate is the label-free geometric rule. The dashed candidate is the end pinch discussed in Section 4. The model sees the candidates drawn in color on the photograph with shuffled numbers, as in Fig. 6.

Because the menu is fixed before the model sees it, the run comes with its own floor and ceiling computed on the same candidates: the expected accuracy of a uniformly random choice, and whether any candidate passes at all. It also yields, as a by-product, a fourth non-learned baseline, the candidate at the centroid closing across the short axis, which unlike System A can express a diagonal grasp.

The prompt was developed on the same 30 training images System C used, two repeats each, and judged on parse rate and on whether the model was reading the glyph as intended. The first version drew each candidate as a thin line with small end ticks. The model frequently chose the candidate running along the object’s long axis while describing a grasp “across the narrowest part”, which could have meant it was reading the line as the finger plates instead of the closing direction. The second version drew explicit pads and arrowheads and described them in the prompt. The behavior did not change (Section 3.6), which suggests that the test result reflects the model’s preference and is unlikely to be an artifact of that particular glyph. Accuracy on the development images was not used to choose between the versions. The second version was frozen, and the test split was called once, five times per image.

After that run, mark size remained as a possible confound. On thin objects the correct marks are drawn with the small opening the labels themselves use, so they are the smallest marks on the image, and a preference for large marks would mimic a preference for the long-axis grasp. Two further runs address it. Both use four candidates at the centroid only, the short-axis direction and three rotations of it, so position is removed and the end-straddling candidates do not exist. In the first, each mark is drawn at its real opening. In the second, the same four candidates with the same scored geometry are drawn at one uniform length, and the prompt gains a single sentence saying the length is not to scale. Each was parse-checked once on the 30 training images and then called on the test split five times per image, under its own sentinel.

2.6 Stratified analyses

Each system’s design predicts something different about how orientation should affect it. System A’s representation cannot express a diagonal grasp, System B’s encoding handles rotation by construction, and System C’s design makes no prediction in either direction.

An image counts as axis-aligned when any one of its labeled grasps sits within 15 degrees of 0 or 90 degrees, and as diagonal only when none does. The rule is deliberately conservative, since a system that can emit only axis-aligned rectangles still has a valid target whenever a single labeled grasp is axis-aligned. It is also looser than the 30 degree scoring tolerance, so a diagonal image whose labeled grasps sit 20 degrees off axis is still reachable by an axis-aligned prediction. Only System A’s direction on this split was predicted before the stratum was scored.

A second split, by the number of grasps labeled per image, was examined as a difficulty proxy, on the expectation that more labeled grasps give a prediction more chances to match.

2.7 Diagnostics and statistical analysis

Two diagnostics are computed for every system from the output of the shared scorer. The failure taxonomy records whether a missed prediction failed the angle criterion, the overlap criterion, or both, or whether no prediction was produced. The center-hull check asks whether each predicted center fell inside the convex hull of an image’s labeled grasp rectangles, the loosest spatial test a prediction can fail.

Per-system accuracies are reported with 95% Wilson intervals, and the paired comparison between Systems A and B uses McNemar’s exact test. System C is tested separately for each of its five runs and the least significant result is reported. Because the 123 test images are grouped within 35 objects, these image-level quantities understate the uncertainty. The primary intervals therefore resample whole objects with replacement, differences between systems are bootstrapped on the same resamples, and an object-clustered permutation test accompanies each difference.

3 Results

3.1 The data split

Table 1. Object-wise split of the Cornell Grasping Dataset. No object appears in more than one split.

Split	Objects	Images
Train	164	620
Validation	35	140
Test	35	123
Total	234	883

The split covers 234 objects across 883 images, divided roughly 70/15/15 by object count. No object appears in more than one split.

Of the 884 frame boundaries examined, 251 (28.4%) fell into the ambiguous band. 139 of those were decided by hand, and the remaining 112 were confident same-object calls accepted without review. Six stayed uncertain after full review. Four carried a lean toward “same object” and were resolved that way, and the other two had no lean. One image from the smaller side of each was dropped instead, accounting for the 883 images retained out of 885.

3.2 Overall accuracy

Table 2. Test accuracy of every system on the same 123 images, with 95% object-clustered intervals. Systems C and D are means over five calls per image.

System	Accuracy	95% CI
A, detector and lookup table (post-hoc corrected)	57.7%	[46.2, 68.2]
A, before the correction	40.7%	—
B, ResNet18, round 1, one seed	79.7%	[70.8, 87.1]
B, ResNet34, round 2, three seeds	84.0%	[73.9, 92.1]
C, GPT-4o coordinates, one call	12.4%	[8.0, 17.4]
C, best of five calls	35.0%	[24.6, 45.5]
D, GPT-4o choosing among 12 candidates	18.7%	[9.5, 29.7]
Label-free short-axis rule	55.3%	[41.8, 68.0]

Table 2 collects the main results, and Fig. 3 shows the three main systems.

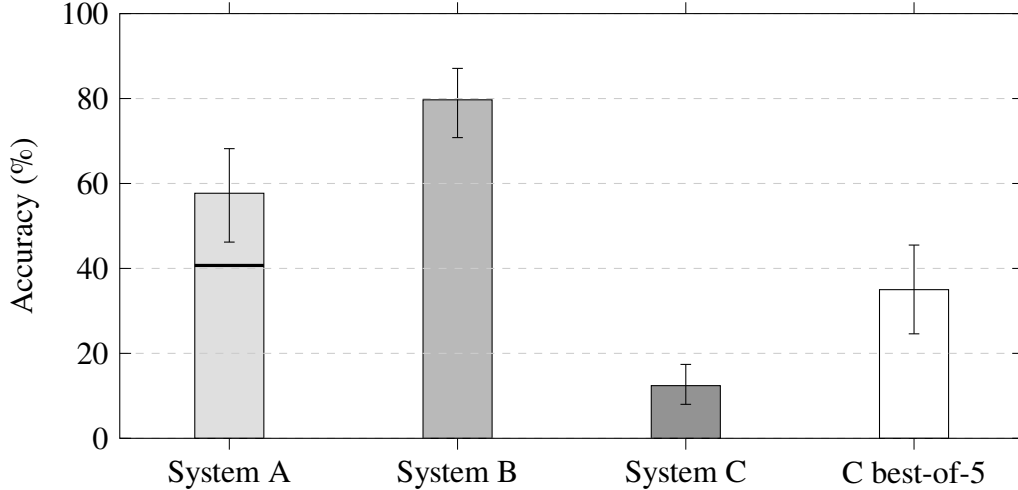


Fig. 3. Test-set accuracy with 95% object-clustered intervals, the primary intervals in this paper. System A 57.7 [46.2, 68.2], System B 79.7 [70.8, 87.1], System C 12.4 [8.0, 17.4], and System C’s best-of-five ceiling 35.0 [24.6, 45.5], which requires five calls per image rather than one. The horizontal rule across the System A bar marks 40.7%, the figure that system scored before the post-hoc geometric guard was added, so the size of that correction is visible here as well as in the text. System B’s bar is the round-1 single-seed figure, and the three-seed round-2 figure is 84.0 [73.9, 92.1], reported in Table 3.

The ordering in Fig. 3 is the paper’s main result. The per-system subsections below give the counts behind each bar.

3.3 System A

System A achieved 57.7% on the test split (71 of 123), with a 95% Wilson interval of [48.9, 66.1]. Of the 52 failed predictions, 13 missed the orientation step alone.

The two paths performed differently. The detector fired on 39.8% of test images (49 of 123). Counting every image where it fired on nothing as a failure gives 22.8% (28 of 123), while accuracy on the images where it did fire was 57.1% (28 of 49). The fallback geometric system attempted 66 images and found no grasp on 8. Accuracy by object category was uneven, with all apples correct (6 of 6) but only 8 of 11 cell phones, none of the cups or bowls, and none of the laptops.

The table was evaluated once at 40.7% before the detector was found to be drawing boxes around background clutter. A geometric guard requiring the box to contain the object’s centroid, plus one constant recalibrated on training, raised the result to 57.7%. Both figures are reported. The detector-only number moved only from 22.0% to 22.8%.

3.4 System B

In round 1, System B’s ResNet18 achieved 79.7% on the test split (98 of 123), with a 95% Wilson interval of [71.7, 85.8]. Its mean angle error of 3.9 degrees is the average, over all 123 test images, of

the smallest angle difference between the prediction and any labeled grasp on that image, regardless of whether the prediction overlapped that grasp or passed the metric. Restricted to the 98 correct predictions and measured against the labeled grasp that the metric matched, the mean angle error is 7.7 degrees. The first figure says the network’s orientation is usually close to some valid orientation on the image. The second says how precisely it orients the grasp it places. System B’s ResNet34 model achieved 70.7% (87 of 123) with an average intersection over union of 0.424 of its grasps with the labeled grasps. The from-scratch model achieved 23.6% (29 of 123) with an average angle error of 24.2 degrees. Parameter counts were 11.3M, 21.4M, and 1.0M respectively.

In round 1, System B’s ResNet18 outperformed ResNet34 on validation as well as test (83.6% versus 77.1%), so the selection held across both splits. The second round below shows that gap was noise. The validation-to-test gap (3.9 points) lies well inside the 6.7 to 11.9 point swings seen between consecutive epochs during training, so it is not evidence of overfitting. The from-scratch network received its own five-point learning-rate sweep on validation, scoring 21.4% at $3e-5$, 27.1% at $1e-4$, 26.4% at $3e-4$, 20.0% at $1e-3$, and 22.9% at $3e-3$. The rate of $1e-4$ was selected.

System B’s ResNet18 model missed 25 of 123 test images. Of those missed images, 4 failed on determining the angle of the object to grasp, while 15 failed on the overlap of the bounding box around the object with any labeled grasp in the image. ResNet34 failed to detect grasps in 36 images, while the from-scratch model missed 94 images. One of those angle-only failures is shown in Fig. 4.

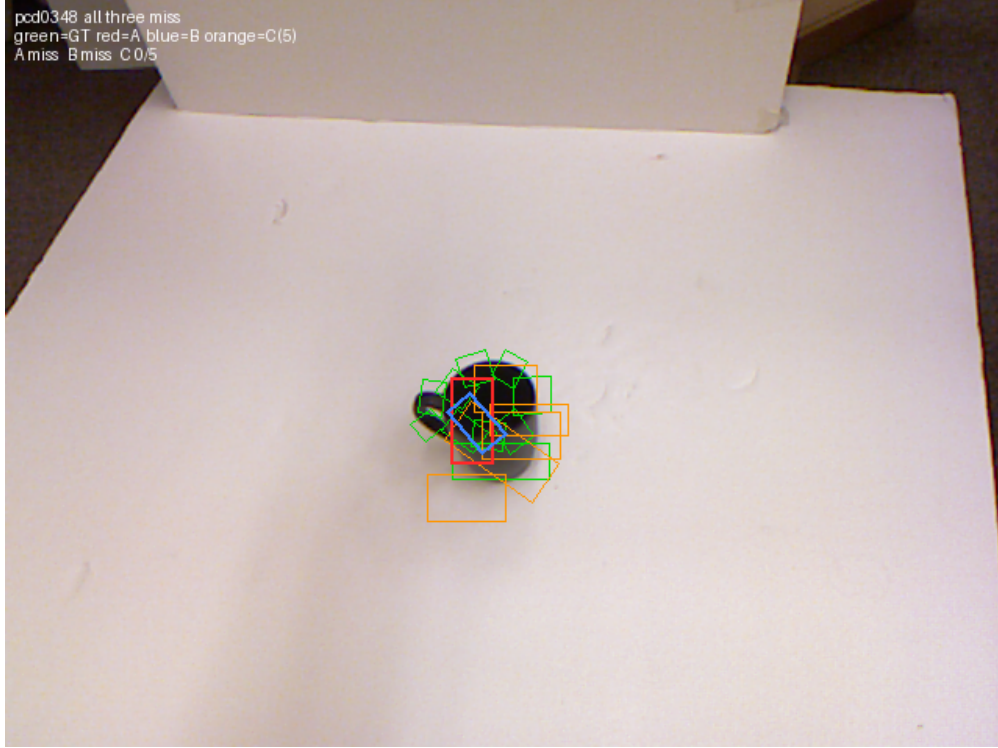


Fig. 4. An orientation-only failure. The predicted grasping rectangle correctly overlaps the object (intersection over union of 0.38) but rotates 47 degrees from any labeled grasp. Of the four ResNet18 angle-only failures (pcd0676, pcd0348, pcd0824, pcd0316), only pcd0348 has a rendered sheet, which is why it was chosen.

The second round, in Table 3, chose ResNet34 by a margin of 0.2 validation points (81.4% against 81.2% mean final weights over three seeds) and used augmentation, which lifted mean validation from 81.4% to 83.3%. On test the three ResNet34 seeds scored 84.6, 83.7, and 83.7%, a mean of 84.0% with a standard deviation of 0.5. ResNet18 under the same recipe scored 83.7, 80.5, and 83.7%, a mean of 82.6%. Re-running the first-round recipe at two new seeds gave validation-selected checkpoints scoring 82.9% and 80.5% on test, so the first round's 79.7% was a low-side sample from a recipe whose seed spread is about 1.6 points at the checkpoint and much larger at the final weights, where the same two runs scored 80.5% and 62.6%. Averaged final weights beat the validation-selected checkpoint of their own run on five of six second-round runs. Test-time augmentation added between 0.0 and 5.7 points across the six runs, about three on average.

Table 3. System B, both rounds, on the same 123 test images. Round 2 numbers are means over three seeds of the final-epoch averaged weights; no checkpoint was selected.

Round	Configuration	Test accuracy
1	ResNet18, seed 42, best validation checkpoint	79.7%
1	ResNet34, seed 42, best validation checkpoint	70.7%
1 recipe	ResNet18, seeds 0 and 1, best validation checkpoint	82.9%, 80.5%
2	ResNet18, three seeds, final weights, with augmentation	82.6% (83.7, 80.5, 83.7)
2	ResNet34, three seeds, final weights, with augmentation	84.0% (84.6, 83.7, 83.7)

Object-clustered, round 2’s 84.0% has an interval of [73.9, 92.1]. It leads System A by 26.3 points [14.8, 38.3], permutation $p = 0.0002$, and System C by 71.7 points [61.3, 80.9]. Its 4.3-point lead over round 1 has an interval of [−5.8, +13.7] and does not exclude zero, so the second round confirms the first-round result with measured uncertainty and does not demonstrate an improvement. Round 1’s nine-point advantage of ResNet18 over ResNet34 is not supported, since the two architectures perform equally within noise.

3.5 System C

System C achieved 12.4% accuracy on test images (76 of 615 calls), with a 95% Wilson interval of [10.0, 15.2]. Each of the five independent runs achieved 13.0%, 14.6%, 13.8%, 10.6%, and 9.8% accuracy. The best-of-five attempts achieved 35.0% (43 of 123). A consensus rule that scores whichever single repeat agrees most closely with the other four, since continuous rectangles cannot be majority-voted directly, reached 12.2%. Neither is the primary figure. Both require either multiple calls per image or knowing the answer in advance. The model returned a parseable reply on 614 of 615 calls (99.8%), and every non-parsing call counts as a miss in the primary figure.

Table 4. Which criterion each prediction missed, computed through one shared scoring path for all three systems.

Outcome	System A (of 123)	System B (of 123)	System C (of 615)
Correct	71 (57.7%)	98 (79.7%)	76 (12.4%)
Angle only	13 (10.6%)	4 (3.3%)	85 (13.8%)
Overlap only	18 (14.6%)	15 (12.2%)	88 (14.3%)
Both	13 (10.6%)	6 (4.9%)	365 (59.3%)
No prediction	8 (6.5%)	0 (0.0%)	1 (0.2%)

The failure taxonomy in Table 4 shows where the three systems differ most. For the geometric check described in Methods, System C’s predicted center fell outside the convex hull containing every labeled grasp on 53.4% of calls (328 of 614). System A showed failures of this type on 7.0%

of the calls in which it attempted a grasp (8 of 115 scored calls), and System B on 6.5% (8 of 123). Table 5 puts the three side by side.

Table 5. Predicted grasp centers falling outside the convex hull of an image’s labeled grasps. This diagnostic is computed across all scored calls, so it overlaps the failure categories in Table 4.

System	Calls scored	Center outside hull
System A	115	8 (7.0%)
System B	123	8 (6.5%)
System C	614	328 (53.4%)

Agreement across System C’s five runs was low. Mean self-agreement was 22.0%, with a mean pairwise angle spread of 5.8 degrees and mean pairwise intersection over union of 0.13. The distribution of how many of five repeats scored correct per image appears in Fig. 5. That puts 81 of 123 images (65.9%) at a fully consistent outcome, leaving 42 where the same model on the same pixels sometimes passed and sometimes did not.

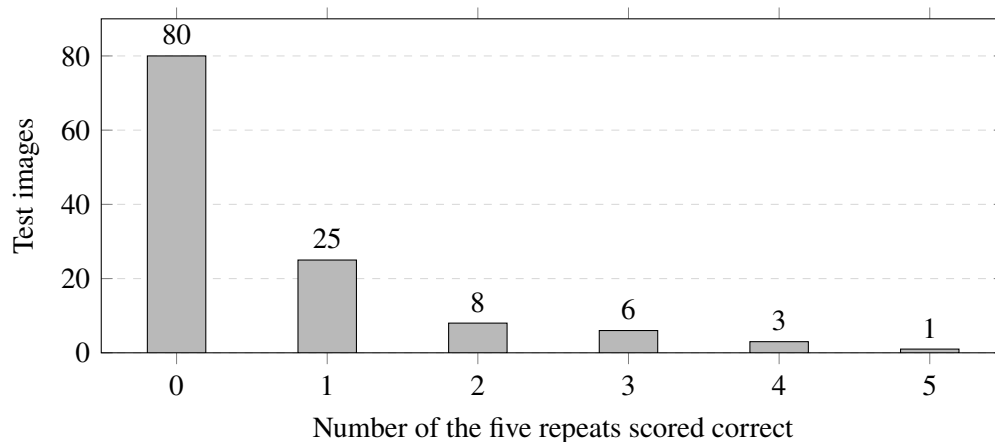


Fig. 5. System C repeat consistency across 123 test images. On 80 images no repeat succeeded and on only one image did all five, leaving 42 images where the same model on the same pixels sometimes passed and sometimes did not.

Self-agreement provides some usable information. Restricting to the 21.1% of images where at least 40% of repeat pairs agreed lifts best-of-five accuracy to 57.7% on those images, against 35.0% at full coverage. The match with System A’s single-prediction 57.7% is coincidental. This result indicates which of several repeated calls to trust and is not comparable with the single-prediction accuracies of Systems A and B.

3.6 System D

System D chose a passing candidate on 18.7% of 615 calls (per repeat 17.1, 17.1, 17.9, 20.3, and 21.1%), with a best-of-five of 39.0%. Every called reply parsed (560 of 560; the other 55 calls

belong to the eleven images with no candidates and count as misses). Table 6 sets that against the floor and ceiling of the same menu, and Fig. 6 shows one test image with the marks the model saw. The paired object-clustered difference between System D and a random choice from the same menu is -1.4 points $[-6.6, +4.0]$. System D’s gain over System C’s coordinate output, 6.3 points $[-1.6, +15.0]$, does not exclude zero either. The geometric rule in the fourth row is statistically level with System A’s post-hoc 57.7% (-2.4 points $[-10.1, +5.4]$).

Table 6. System D against the floor and ceiling of its own candidate menu, and against System C, on the same 123 test images. Intervals are object-clustered.

	Accuracy	95% CI
System D, one call per image	18.7%	[9.5, 29.7]
Random choice among candidates	20.1%	[14.6, 26.6]
Any candidate passes (ceiling)	79.7%	[67.2, 90.0]
Geometric rule alone (candidate 0)	55.3%	[41.8, 68.0]
System C, one call per image	12.4%	[8.0, 17.4]

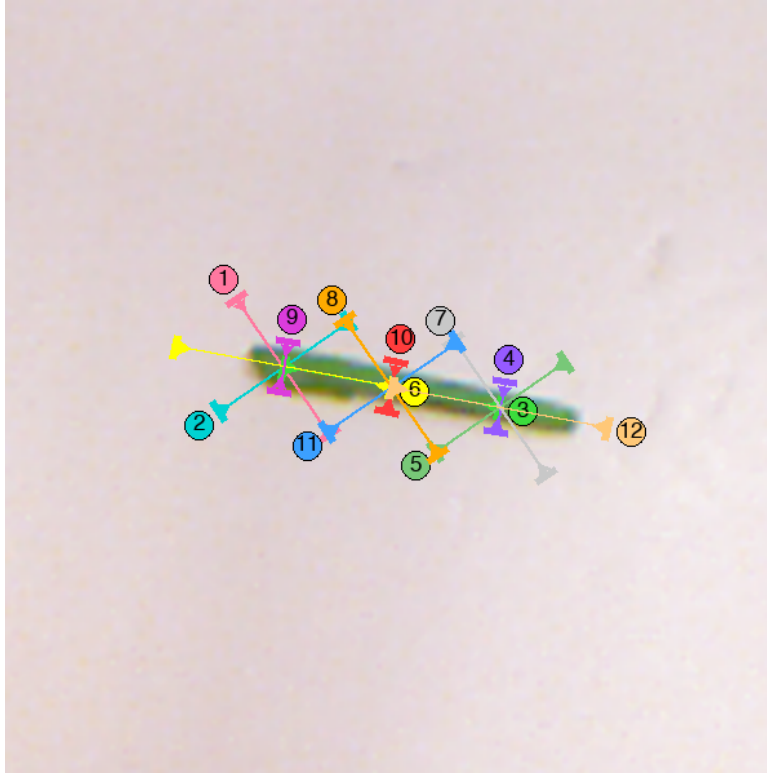


Fig. 6. The marked image for one test object as System D saw it. The model chose mark 6, the yellow end pinch along the bean, on all five repeats, describing it as gripping “the central narrow part of the bean”. Mark 10 crosses the bean at its centre and passes the metric.

Self-agreement across the five repeats, measured on the physical candidate with the num-

bering re-shuffled each call, was 44.3%, twice System C’s. The preference it agreed on was for the candidate whose jaws travel along the object’s long axis near one end: 342 of 560 choices for that single candidate, 391 for that orientation at any position. That candidate type passes on 17.0% of test images. The centroid candidate closing across the short axis passes on 60.7% and was chosen 34 times. The model’s stated confidence was “high” on 556 of 560 calls, of which 21% were correct. The same preference had appeared during prompt development on the 30 training images, in 34 of 58 calls with the first glyph and 41 of 58 with the redesigned one.

The two follow-up menus are in Table 7. On the four-candidate menu drawn at real size the model scored 26.5% against a 24.6% random floor. With every mark the same size it scored 22.3% against the same floor. Neither paired difference excludes zero (+1.9 [−2.2, +6.3] and −2.3 [−6.3, +1.7]), nor does the difference between the two menus (+4.2 [−0.5, +9.4]). The sparse menu’s 7.8 point lead over the full menu excludes zero, but its floor is 4.5 points higher, so the gain is attributable to the easier menu. On the uniform-size menu the model chose the long-axis direction in 45% of calls against a 25% chance rate, and the short-axis candidate, which passes on 55% of images, in 17%.

Table 7. System D on three candidate menus, same model, same 123 test images, five repeats each. “Long axis” is the share of parsed calls choosing the orientation whose jaws travel along the object’s long axis; chance is 25% on the four-candidate menus.

Menu	Marks	Accuracy	Random floor	Ceiling	Long axis chosen
Full, three positions	12	18.7%	20.1%	79.7%	70%
Centroid only, real size	4	26.5%	24.6%	65.9%	41%
Centroid only, uniform size	4	22.3%	24.6%	65.9%	45%

3.7 Cross-system comparison

System A against System B is significant under McNemar’s exact test ($p = 1.4\text{e-}4$), the 38-versus-11 split described below. System C was tested separately against each of its five runs rather than pooled, and the least significant of the five is reported, $p = 5\text{e-}12$ against System A and $p = 1.6\text{e-}20$ against System B.

Every interval quoted so far treats images, or in System C’s case individual calls, as independent. They are not, because the 123 test images sit inside 35 objects. Resampling whole objects with replacement widens all of them. System A becomes [46.2, 68.2], System B [70.8, 87.1], System C [8.0, 17.4], and System C’s best-of-five [24.6, 45.5]. Checking whether two such intervals overlap is a weak test, so the differences were bootstrapped directly on the same resamples. System B leads System A by 22.0 points [9.7, 35.1], System A leads System C’s best-of-five by 22.8 points [9.2, 35.6], and System A leads System C at one call per image by 45.4 points [33.8, 55.6]. None of those intervals contains zero, and an object-clustered permutation test agrees at $p = 0.003$, 0.006 , and $5\text{e-}5$. The ordering of the three systems survives the change of sampling unit. The Wilson intervals reported per system above are image-level and should be read as the narrower of the two.

Systems A and B agreed on 74 of 123 images, both correct on 60 and both wrong on 14. Of the 49 they disagreed on, System B alone was correct on 38 and System A alone on 11. Counting System C as correct whenever any of its five repeats passed, all three systems succeeded together on 22 images and failed together on 10. Fig. 7 shows one such image with all three systems overlaid.

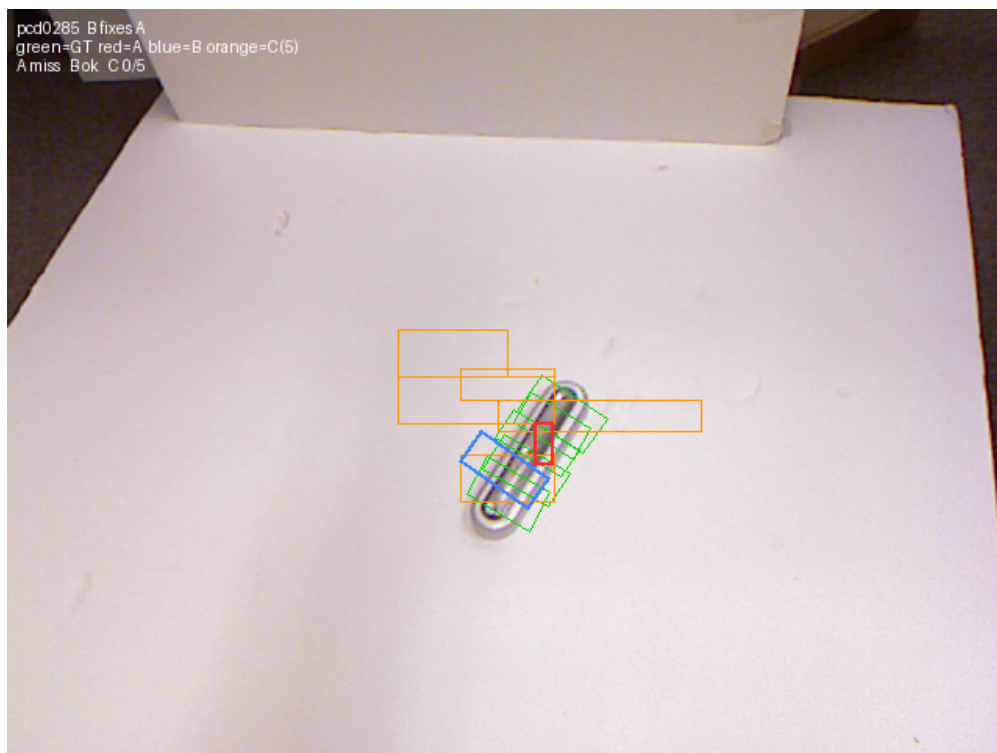


Fig. 7. All three systems on one image. Labeled ground-truth grasps are green, System A red, System B blue, and System C’s five repeats orange.

3.8 Orientation and label-count strata

Fig. 8 splits the test set by orientation. Relative to the 103 axis-aligned images, System A dropped 21.2 points on the 20 diagonal images, System B gained 12.3 points, and System C moved 2.0 points. System A does not fall to zero on the diagonal stratum because the 15 degree stratum rule is tighter than the 30 degree scoring tolerance, so some diagonal images remain reachable by an axis-aligned rectangle.

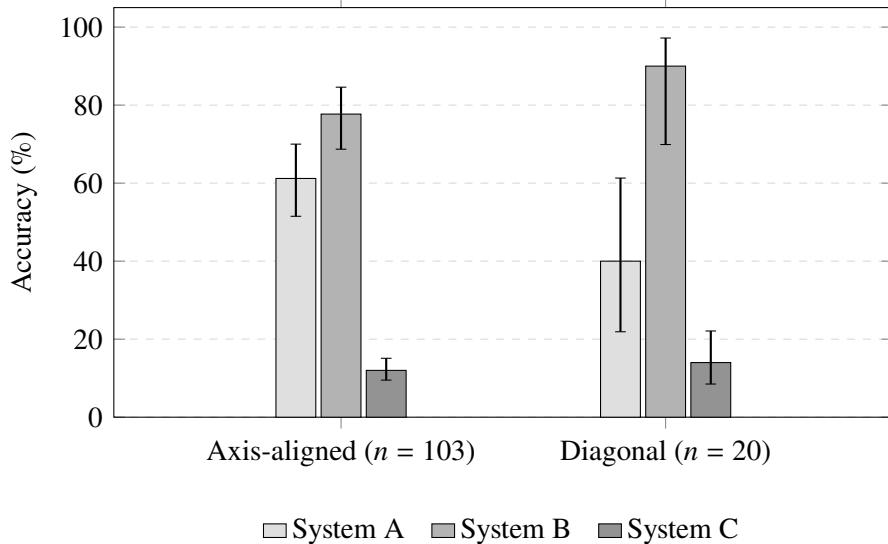


Fig. 8. Accuracy by orientation stratum with 95% Wilson intervals. Axis-aligned, 103 images: System A 61.2 [51.5, 70.0], System B 77.7 [68.7, 84.6], System C 12.0 [9.5, 15.1]. Diagonal, 20 images: System A 40.0 [21.9, 61.3], System B 90.0 [69.9, 97.2], System C 14.0 [8.5, 22.1]. These are image-level rather than object-clustered, so they are the narrower reading. The three systems move in three directions, but the diagonal intervals are already wide enough that only System A’s drop, which was predicted from its representation beforehand, should carry weight.

All three systems performed worst on the images with the most labeled grasps (8 to 25), not the fewest, which is the opposite of what a more-labels-means-easier-match hypothesis would predict. In that group, System A scored 48.0%, System B 64.0%, and System C 11.2%.

4 Discussion

4.1 From the coordinate-binding hypothesis to System D

The most distinctive feature of System C’s errors is their location. Most of its calls failed the angle and the overlap criterion together (Table 4), and its predicted center fell outside the convex hull of the labeled grasps far more often than either other system’s (Table 5). The comparison is not fully even, since System B was trained on this distribution and System A’s center comes from a detected box or a segmentation, so both almost always place the center on the object, whereas System C is free to place a center anywhere in the image. In a sample of failures, the model’s written explanation frequently named a real part of the object while the coordinates in the same reply fell elsewhere in the image. That reading was informal and not a blinded coding study with a fixed rubric, so it is reported only as the observation that motivated a hypothesis, and the frequency of the pattern was not measured. Wang et al. [14] document the same disconnect in an unrelated domain, with models describing a correct solution to a visual puzzle in text, then clicking hundreds of pixels away from it (their section 5.3.2). Together these suggested that GPT-4o could see where to grasp but could not bind that choice to pixel coordinates.

This explanation predicts that removing System C’s coordinate output should help if the problem is binding reasoning to coordinates, and should make no difference if the problem is perception of the object. System D ran that test, and the prediction was not borne out. Given a menu of twelve candidates, GPT-4o chose a passing grasp no more often than a uniformly random choice from the same menu, and its gain over its own coordinate output does not exclude zero (Table 6). Two published results had seemed consistent with the binding reading, though neither was designed as a test of it. They remain relevant because they show what a menu-based design can do when the menu and the model are matched. Set-of-Mark lets zero-shot GPT-4V beat a fine-tuned specialist on RefCOCOg purely by replacing coordinate output with region selection [9]. VLAD-Grasp scores far above System C on this same dataset, training-free, by having the model draw the grasp instead of stating it [10].

VLAD-Grasp is the published result most relevant to System C, but two differences prevent a direct comparison with 12.4%. First, it does not only replace a coordinate with a picture. It then predicts depth, segments the object, and aligns 3D point clouds to recover a pose, a pipeline System C has no equivalent to. Second, it counts a grasp correct on overlap alone, with no angle requirement, and reports 91.4% under that looser test on a set of 70 unseen Cornell objects. That the same evaluation scores GR-ConvNet at 72.1%, against the 97.7% GR-ConvNet reports for itself, shows the two protocols are not directly comparable. The dropped angle criterion is what matters here, since orientation is the variable this paper’s results depend on most. The model also differs, since the current revision of that paper names GPT-5 as its primary model, so its result describes vision-language models without coordinate output in general and cannot be attributed to GPT-4o.

Even with that caveat, the contrast between VLAD-Grasp and System D is informative in a way System C alone was not. Both remove the coordinate output, but only VLAD-Grasp succeeds, and it also adds depth, segmentation, and geometric pose recovery, so the geometry contributes a substantial share of that pipeline’s performance. On the full menu, System D most often chose the candidate whose jaws travel along the object’s long axis near one end, a type of grasp that rarely passes the metric, and seldom chose the centroid candidate closing across the short axis, which passes most often (Section 3.6). Its reasoning for the preferred choice was typically that it “grips the narrow top edge near the center of mass”. From a top-down photograph, two pads straddling the end of an elongated object appear to pinch a narrow part. In three dimensions the second finger would land on top of the object. The model reported high confidence on nearly every call, whether or not the choice was correct. The failure that the center-hull diagnostic exposed in System C is therefore not explained by the coordinate output alone, and it is more consistent with a failure of grasp reasoning from a single 2-D view. One rendering explanation, mark size, was tested and ruled out. With four candidates at the centroid and every mark drawn at the same size, the model still chose the long-axis direction well above the chance rate and the short-axis candidate below it (Table 7). The preference is weaker without end-straddling candidates, so part of the full-menu result was the end-pinch illusion specifically and part is a general preference for closing along the long axis.

These controls do not establish that the model reads the marks as intended. The glyph redesign and the uniform-size menu address two specific rendering confounds, but the model could

still misjudge the direction in which a drawn candidate closes, and its description of an end pinch as a grip on a narrow part is compatible with that. A direct control would ask the model a purely perceptual question about the same marks, for example which candidate closes across the object’s short axis. That question has a known answer from the candidate generator and needs no grasp label. High accuracy on that question would indicate that the failure lies in the grasp choice, whereas low accuracy would implicate mark interpretation. The control has not been run, so the claim made here is limited to the statement that removing the coordinate output is not sufficient to recover performance. The single-view setting may also matter. Zhou et al. study vision-language models that must combine several agents’ egocentric views into one spatial understanding, and improve that ability with supervised and reinforcement fine-tuning [15]. Whether a second viewpoint, or depth, would change the grasp that GPT-4o chooses was not tested here.

4.2 Comparison with published regression results

System B’s accuracy can appear low next to two published Cornell results, but neither is an even comparison. GR-ConvNet reports 97.7% image-wise and 96.6% object-wise on an extended 1,035-image version of Cornell [5], and only the object-wise figure is on the same footing as anything here. Redmon and Angelova report 84.9% object-wise [3]. GR-ConvNet is a pixel-wise method that predicts a grasp at every pixel in a single pass, whereas System B pools a backbone’s features and regresses one rectangle for the whole image, the same design Redmon and Angelova published, whose 84.9% object-wise score is the fairer target, as laid out in Table 8. The remaining differences favor the published system, since Redmon and Angelova used RGB-D and roughly 3,000 augmented examples per image, while System B used RGB only and 620 real training images.

Table 8. System B compared with Redmon and Angelova’s Direct Regression, the closest published architecture.

	Redmon and Angelova (2015)	System B
Architecture	Global regression	Global regression
Orientation encoding	sin/cos of twice the angle	sin/cos of twice the angle
Input	RGB-D	RGB only
Training data	~3,000 augmented per image	620 real images
Backbone	AlexNet	ResNet18, then ResNet34
Object-wise accuracy	84.9%	79.7% (round 1), 84.0% (round 2)

4.3 Orientation and label count

The orientation split produced the pattern each design predicts (Section 3.8). Both effects are expected, since an axis-aligned box cannot represent a diagonal grasp and the sine and cosine encoding was designed to handle rotation. The value of the split is as a diagnostic, a use for which no precedent was found. It checks whether each system’s accuracy rises or falls along an axis defined by

the dataset’s own grasp-angle annotations, instead of reporting a single train and test generalization number.

The diagonal stratum holds only 20 images, and the 95% intervals around System B’s and System C’s per-stratum accuracies are wide enough that their point changes there are not reliably distinguishable from noise. Only System A’s drop was predicted from its representation before the stratum was scored. Predicting a direction in advance is not the same as confirming an effect, and no interaction test between system and stratum was run, so this analysis remains exploratory. System C’s near-flat response is what a system with no working orientation signal would produce.

The label-count split did not behave as a difficulty proxy. All three systems performed worst on the images with the most labeled grasps. The likely explanation is that annotators labeled more grasps on objects that afford more of them, so the count reflects object complexity more than the leniency of the metric.

4.4 What the comparison adds

None of the individual observations is new. The sine and cosine encoding dates to 2015, learned models have outperformed rule-based baselines on Cornell for a decade, and imprecise vision-language coordinates have been documented elsewhere. The contribution of this experiment is to quantify the last weakness on a standard benchmark against two baselines, and then to test the most natural account of it and find it insufficient. A side result of the menu is a rule that closes across the segmented object’s short axis at its centroid, with no learning and no detector. It is statistically level with System A’s post-hoc figure (Table 6), and it is the only label-free baseline in this study that can express a diagonal grasp.

The second training round also corrected an earlier interpretation. The first round’s nine-point gap between ResNet18 and ResNet34 had been attributed to the smaller network suiting a small dataset. Three seeds per architecture under a stable recipe placed them within 1.4 points of each other. A single training run on 620 images, with a checkpoint chosen from a hundred noisy validation epochs, does not provide evidence about architecture.

A one-million-parameter network trained from scratch did not learn the task adequately, reaching 23.6% after a learning-rate sweep confirmed that this was not a tuning artifact. It still exceeded GPT-4o (System C) by 11 points, which was not anticipated.

Three defects in the evaluation code were caught before any result was reported, each of which would have produced a plausible but wrong number. Each was found when an automated result did not match an independent calculation, and neither value was used until the disagreement was explained.

5 Limitations

This study used a single split of the dataset, whereas most published Cornell results use five-fold cross-validation. Seven of the 35 object groups in the test split contain only a single image of the

object, so the results for those objects are imprecise. The test split was opened twice for System B, once per round, and the second round’s selection rules were fixed beforehand so that the second opening could not steer any choice. The round-1 numbers rest on one seed each and the round-2 numbers on three. Jacquard, considered as a second dataset, was not available during this study, so generalization across datasets was not tested. An early formatting check on roughly 550 grasp rectangles likely included a small number of images that were later assigned to the test split. No threshold, constant, or lookup entry was derived from that check.

The three systems did not receive comparable engineering effort, which weakens any reading of this as a fully controlled comparison. System B received three architectures, a five-point learning-rate sweep, a tuned loss weight, and validation-based model selection. System C received one prompt, frozen after development on 30 training images and never revised. Freezing it protected test-set integrity, but prompt design is the vision-language equivalent of architecture search, and architecture search was run for one system and not the other. The 12.4% figure therefore measures what a single, un-iterated prompt achieves and should not be read as a ceiling for zero-shot GPT-4o, although the gap to System B is wide enough that prompt iteration is unlikely to reverse the ordering. System D shares this limit and has a further one, since all three menus came from one candidate generator and were drawn in one glyph style on a zoomed crop. A different mark style, or marks placed by a segmentation the model itself produced, was not tested. No control measured whether the model reads the closing direction of a mark correctly, so mark interpretation remains a possible contributor to System D’s result.

Cornell is public and widely mirrored, so GPT-4o may have been exposed to it during training. When asked to name the source of ten test images from different object categories, the model recognized none of them. That is weak evidence against contamination. Contamination could only have raised System C’s score, which is already the lowest of the three, so its ranking holds even in the worst case.

Every image-level and call-level interval and significance test reported here treats its unit of analysis as independent, which does not hold here. System C’s interval is computed over 615 calls, but those calls are five repeats of 123 images, so the interval is narrower than the data support. The image-level intervals and McNemar tests for Systems A and B have the same problem one level up, since the 123 test images are grouped within 35 objects and images of one object are not independent of each other. The stated population is unseen objects, so the correct sampling unit is the object. The object-clustered intervals and differences reported in the results section are the primary ones, and the per-system Wilson intervals are image-level and should be read as lower bounds on the uncertainty. Weighting each object equally rather than by how many images it contributes lowers every system, System A to 51.3%, System B to 75.4%, and System C to 10.5%, because test object groups range from one image to nine, but the ordering of the systems does not change.

The orientation analysis rests on only 20 diagonal images. The vision-language results come from one model and one way of posing the question, so they describe that setup and should not be extended to vision-language models in general.

All systems received RGB images only, and no depth information was used.

6 Conclusion

A grasp decision can be written as rules, learned from labeled data, or requested from a general-purpose vision-language model. These options differ greatly in what they cost to build, and they are seldom measured side by side. This paper built one system of each kind and scored all of them with one implementation of the Cornell rectangle metric, on the same 123 held-out images of 35 objects that none of them had seen, with uncertainty computed at the level of the object.

The ordering was consistent across every analysis. A fine-tuned ResNet34 reached 84.0% averaged over three seeds (a single-seed ResNet18 reached 79.7% in the first round), a detector with a lookup table reached a post-hoc-corrected 57.7%, and zero-shot GPT-4o reached 12.4%. The manner in which each system failed is more informative than the accuracy figures. The rule-based system usually finds the object but can only place an axis-aligned rectangle, so it loses accuracy where the labeled grasps run diagonally. The learned system orients well, to within about eight degrees on its correct predictions, and most of its remaining error is placement. GPT-4o fails at an earlier stage than either. More than half of its predicted centers fall outside the region spanned by all labeled grasps, often while its own text describes a sensible part of the object.

That pattern suggested a model that sees the right grasp and cannot express it in pixels. The marked-candidate experiment tested that account and did not support it. Given numbered candidates to choose from, with a ceiling of 79.7%, the same model performed at the random floor of the menu on three menus, and it consistently preferred a pinch along the object’s long axis, which resembles a grip on a narrow part only when viewed from above. Removing the coordinate output is therefore not sufficient to recover performance. The evidence points to the grasp choice itself, although the model’s reading of the drawn marks has not been measured separately.

These results have three practical implications. First, a hosted vision-language model prompted once for a grasp pose is not yet a substitute for even a simple heuristic, which is consistent with published systems that use such models to choose an object or a part and leave the pose to geometry. Second, a label-free rule that closes across the short axis of the segmented object matched the detector-based baseline and can express diagonal grasps, which makes it a stronger low-cost baseline than a lookup table. Third, results on a benchmark of this size are sensitive to the training seed and to the unit of analysis. A nine-point gap between two architectures disappeared under three seeds, and every interval widened once images were clustered by object, so multiple seeds and object-level clustering are recommended for work at this scale. The next steps are a perceptual control on the candidate marks, depth or multi-view input, newer models, and a second dataset.

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